THEJUS MAHAJAN PH.D.

Computational Scientist | Clinical Data Pipelines · Biostatistics · Bioinformatics

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Google Scholar

PROFILE

- Computational Scientist (PhD, Post-doc) with experience in clinical data pipeline engineering, biostatistics, and bioinformatics.
- Core skills: R (tidyverse, Shiny, tidymodels), Python (JAX, Pandas), SQL, Git.
- HealthTwiSt GmbH (Praxisphase): refactored a production R pipeline for the DeGIR quality registry (143K+ patient records, ~300 clinics), built an R Shiny dashboard, created the degirtools R package.
- Available from mid-April 2026.
- Open to: Data Science, Biostatistics, Bioinformatics, Clinical Data Analysis, Research Software Engineering; open to relocation within Germany.

PROFESSIONAL EXPERIENCE

Clinical Data Engineering — Praxisphase (Weiterbildung ABI-2025-3)

HealthTwiSt GmbH

02/2026 - 04/2026 · Berlin, Germany

Clinical data research and development under Dr. Andreas Busjahn

- Refactored a monolithic 1,834-line R data pipeline (tidyverse) for Germany's interventional radiology quality registry (DeGIR); 143,000+ patient records from ~300 clinics. Reduced to 1,349 lines (−26.5%) with byte-identical output verified via `identical()`.
- Externalised 257 hard-coded correction rules into 8 configuration CSV files, enabling non-programmer editing of business rules.
- Replaced 360 deprecated function calls across 4 pipeline files, modernising the codebase to current tidyverse standards.
- Created the `degirtools` R package (9 exported functions, roxygen2 docs, `devtools::check()` passing) by extracting reusable logic from a 767-line helper file.
- Discovered 2 pre-existing bugs through systematic code analysis; documented both and preserved original behaviour for supervisor review — demonstrating production-code discipline.
- Built an interactive R Shiny dashboard (6 modules: overview, interventions, complications, radiation doses, success rates, about) using plotly, DT, and bslib. Deployed to shinyapps.io with GDPR-safe synthetic data.
- Wrote Tidymodels teaching materials (Quarto/GitHub) replacing the legacy caret framework for ML workflows in R.

Continuing Education – Bioinformatics and Biostatistics

CQ Beratung + Bildung GmbH

08/2025 - 02/2026 · Berlin, Germany

Applied bioinformatics and biostatistics

- Programming methodology: Shell (BASH), Python (object-oriented), R.
- Bioinformatics databases (NCBI, Biopython, SQL) and sequence analysis.
- NGS analysis pipelines (Next-Generation Sequencing; Nextflow, Galaxy); structural bioinformatics.
- Biostatistics (R/Bioconductor): ANOVA, PCA, HCA, multivariate analysis, Markov chains.
- Capstone: Built a reproducible Nextflow (DSL2) pipeline for Hepatitis Delta Virus sequence analysis — automated NCBI download, MAFFT alignment, and trimAl trimming with Conda-managed dependencies (GitHub).

Guest Scientist (parallel to continuing education from 08/2025)

Helmholtz-Zentrum Hereon, Ecosystem Modelling

KEY STRENGTHS

Clinical Data Pipeline Engineering
Refactored a production data pipeline (1,834 lines, 143K records, ~300 clinics) with 26.5% code reduction. Built degirtools R package. Byte-level verification after every change via `identical()`.

Biostatistics & Machine Learning
Proficient in multivariate analysis (PCA, ANOVA), clinical outcome analysis, developing machine learning workflows using tidymodels, and cross-validation.

Bioinformatics & Reproducible Research
Proficient in NGS data analysis, building reproducible pipelines with Nextflow (DSL2), Bioconductor workflows, and ensuring GDPR-compliant data handling.

TECHNICAL SKILLS

Programming
R (tidyverse, Shiny, tidymodels, Bioconductor), Python (JAX, Pandas, NumPy), SQL, Fortran, C++, BASH

Clinical Data & Biostatistics
Medical registry pipelines, GDPR/DSGVO compliance, multivariate analysis, PCA, ANOVA, survival analysis concepts, clinical endpoints

Data Engineering
Pipeline refactoring, config-driven ETL, `identical()` verification, R Shiny dashboards, large datasets (HDF5, NetCDF)

Bioinformatics
NGS analysis, Nextflow (DSL2), sequence analysis, Bioconductor

Tools
Git/GitHub, Linux/Unix, HPC clusters, Conda, LaTeX, Docker concepts

LANGUAGES

English — Fluent (C1)

German — B1 (Goethe-Zertifikat), B2 exam scheduled

French — Basic knowledge

05/2025 - 10/2025 Geesthacht, Germany

Modelling evolutionary processes in marine ecosystems; simulation of long-term environmental impacts using advanced computational methods. Finalised the computational modelling framework and prepared the simulation data for peer-reviewed publication.

Job Search & German Language Course

02/2025 - 04/2025 Hamburg, Germany

Active job search and German language acquisition (Goethe-Zertifikat B1).

Post-doctoral Scientist

Universität Hamburg, Institute for Marine Ecosystem and Fishery Science

08/2021 - 01/2025 Hamburg, Germany

Marine ecosystem modelling

- *Parental leave from 01/2024 to 12/2024.*
- Model development: Development of the “Cyanobacteria Life Cycle” (CLC) model within the ERGOM framework for climate warming projections.
- High-performance computing (HPC): Translation of a legacy Fortran model into a high-performance Python engine (**Google JAX**, TPU/GPU parallelisation). Processing of large NetCDF datasets in Linux HPC environment.

Career Transition

04/2021 - 07/2021 Hamburg, Germany

Relocation from India to Germany and active job search.

Physics Tutor & Chess Coach — Freelance / Tutorwaves Solutions Inc., Kerala, India · 10/2018 – 03/2021

Doctoral Researcher in Astrochemistry

Université Paris-Saclay

10/2015 - 09/2018 Orsay, France

Conducting atomic collision experiments, data analysis, and results modelling.

- Acquisition and processing of terabytes of raw data from tandem particle accelerator experiments.
- C++ code optimisation: improvement of existing algorithms for noise suppression, signal smoothing, and conversion of raw data into multi-dimensional structures.
- Development of a molecular fragmentation model for predicting chemical reactions in space (contribution to the KIDA database).

Malayalam

Native

Tamil

Fluent

Hindi

Basic knowledge

COURSES & TRAINING

High Performance Computing (HPC) Training
Jülich Supercomputing **Center (JSC)**, Forschungszentrum Jülich

MITx: 7.00x Introduction to Biology – The Secret of Life
Comprehensive coursework in genetics, biochemistry, and molecular biology.

PUBLICATIONS (ASTROCHEMISTRY / PHYSICS, 2018–2020)

T. Mahajan et al. *J. Phys.: Conf. Ser.*, 1412:142026, 2020.

T. Idbarkach et al. *Astron. Astrophys.*, 628:A75, 2019.

T. Mahajan et al. *J. Phys. B*, 2019.

T. Idbarkach et al. *J. Phys. B*, 51(24):245201, 2018.

T. Idbarkach et al. *J. Phys.: Conf. Ser.*, 1412(11):112028, 2020.

Google Scholar Profile

EDUCATION

Ph.D. in Astrochemistry

Université Paris-Saclay, Institute of Molecular Sciences (ISMO)

10/2015 - 09/2018 Orsay, France

Supervisor: Dr. Karine Béroff. Thesis: *Excitation and fragmentation of C_nN^+ (n=1–3) in collisions with He atoms at intermediate velocity.*

M.Sc. in Physics

National Institute of Technology Calicut, India

07/2012 - 12/2014 Calicut, India

First Class with Distinction (CGPA 8.71/10).

Research Project – Theoretical Atomic Physics

IIT Mandi (under Dr. Hari Varma)

02/2015 - 08/2015 Mandi, India

Theoretical atomic physics; partly remote.

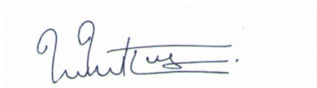
B.Sc. in Physics

University of Calicut, India

06/2009 - 04/2012 Thrissur, India

Grade: B+ (CGPA 3.49/4.0).

Hamburg, 18 April 2026



Dr. Thejus Mahajan